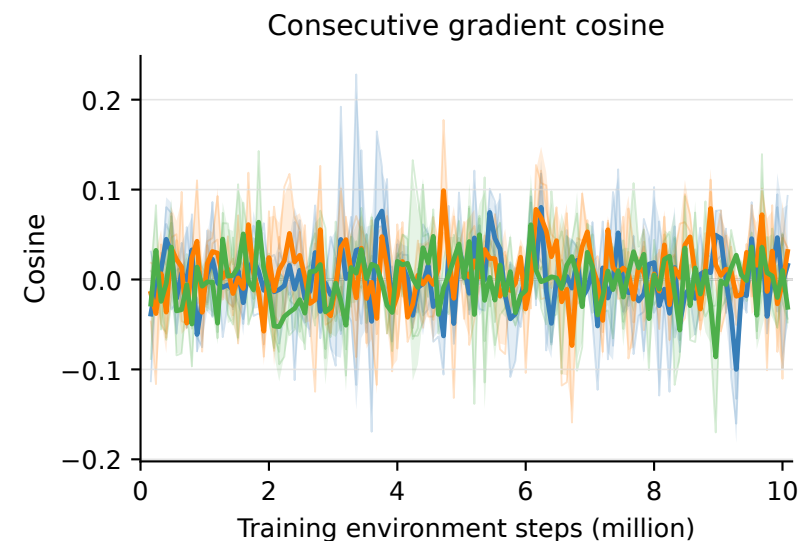
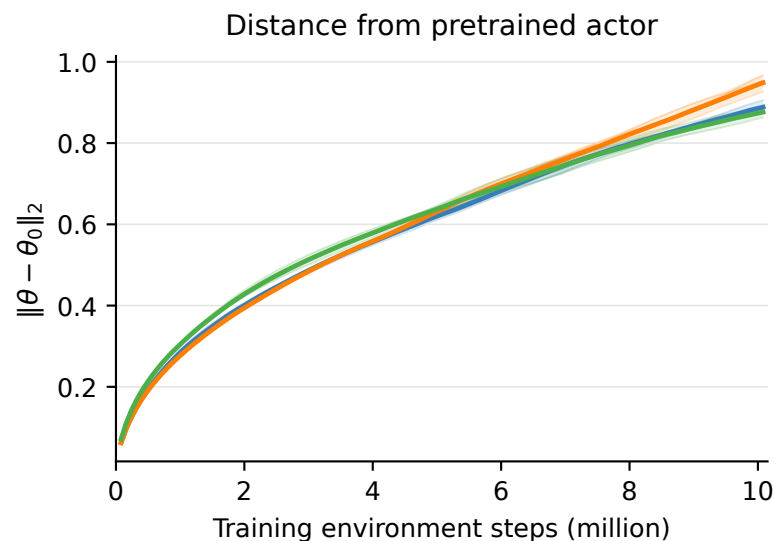


Stage 3: update diagnostics across completed wave 1

Nine conditions · seeds 0-2 · mean $\pm$ sample SD · faint individual seeds · no smoothing

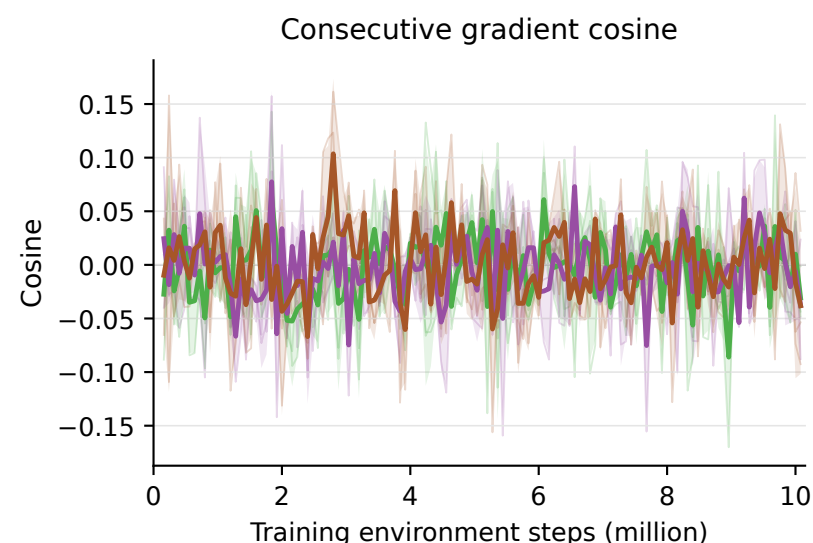
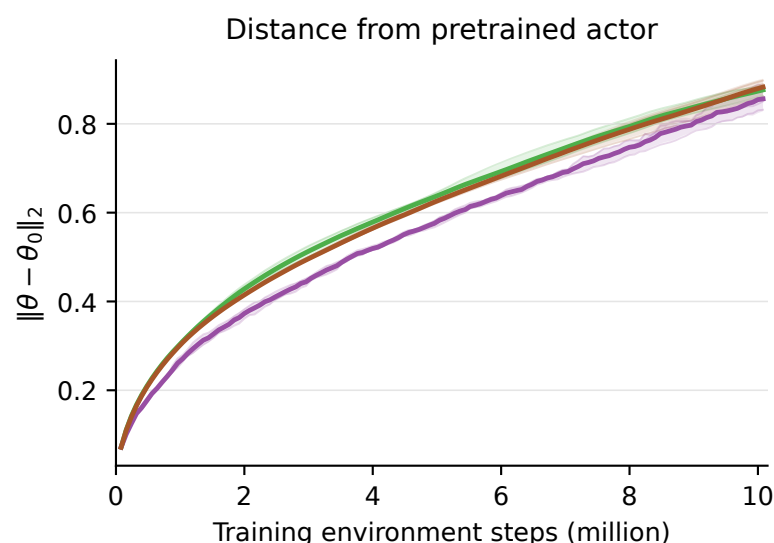
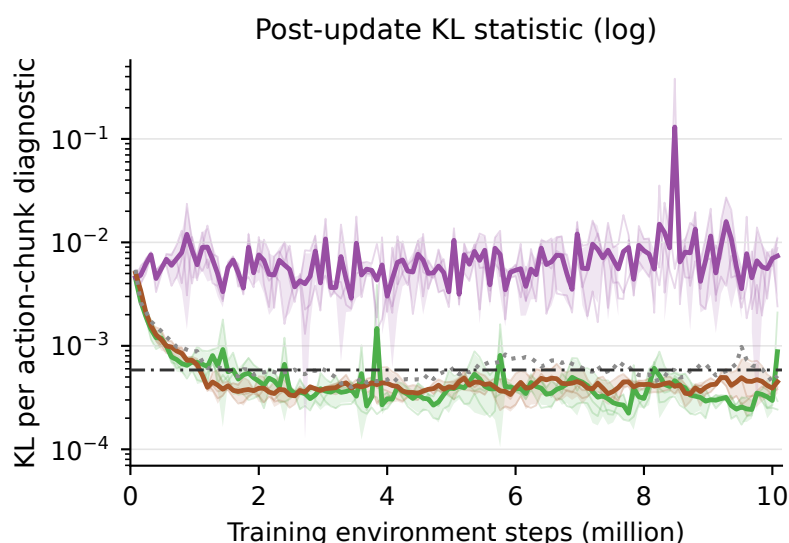
Estimator ladder under AdamW

E0 E1 E2 Stage 2 NC4 1e-4 Calibration KL*



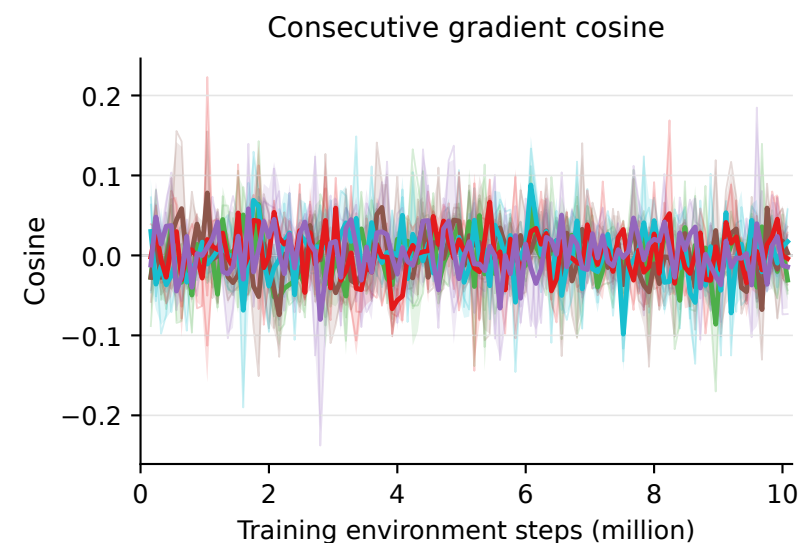
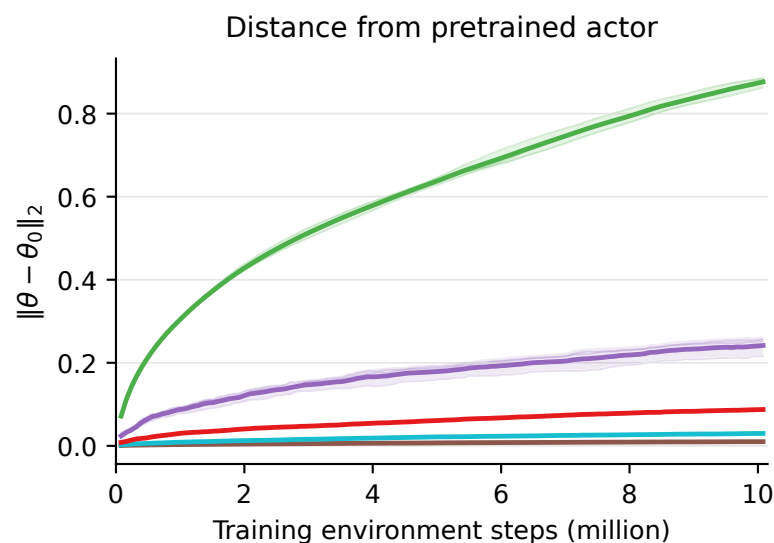
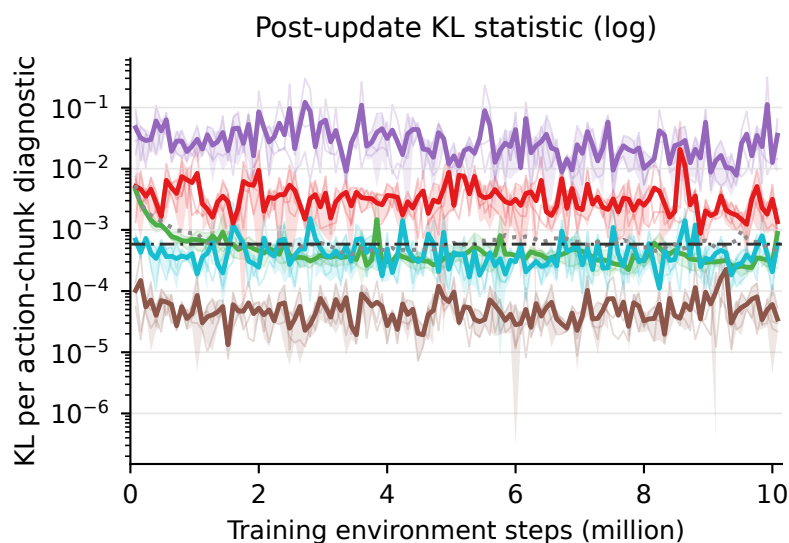
E2 under AdamW: momentum and baseline

E2 E2_beta0 E2_nobase Stage 2 NC4 1e-4 Calibration KL*



E2 estimator: AdamW and the SGD step-size sweep

E2 SGD03 SGD1 SGD3 SGD10 Stage 2 NC4 1e-4 Calibration KL*



First-update gradient cosines are undefined. Gray reference: NC4 KL only; dash-dot: calibration target.